Notes: Bourlès, Bramoullé, and Perez-Richet (Econometrica, 2017)

Altruism in Networks

Contents

1 Big picture

- **Question.** How do private transfers work when altruism is organized through a network rather than a fully connected family or dynasty?
- **Core environment.** Agents have initial incomes and can make bilateral financial transfers. They care about their own private utility and about the private well-being of network neighbors. The altruism network determines who may be supported and how strongly.
- **Why networks matter.** With two agents or a complete family, altruism mainly creates direct redistribution from richer to poorer people. In a network, a transfer may pass through intermediaries, and distant agents can affect each other through chains of altruistic links.
- **Main theoretical result.** Nash equilibria of the transfer game solve a concave potential maximization problem. This converts a seemingly complicated strategic game into a tractable convex-optimization problem.
- **Key economic object.** A transfer from i to j is associated with a virtual cost

  $$c_{ij} = -\ln(\alpha_{ij}),$$

  where α_{ij} is the reduced-form altruism weight. Stronger altruism means lower transfer cost.
- **Equilibrium geometry.** Equilibrium transfers are acyclic and, generically, form a forest. Money flows through least-cost paths of the altruism network. This is the network analogue of saying that transfers choose the strongest available altruistic routes.
- **Intermediaries.** Transfer intermediaries arise when money flows from a rich agent to a poorer agent through another person. The paper shows that intermediaries are tied to intransitivity of altruism: if indirect concern is stronger than direct concern, money may travel indirectly.
- **Neutrality result.** Local neutrality of redistribution does not require the whole altruism network to be connected. It requires redistribution to stay within connected components of the *equilibrium transfer network*. These components behave like income-pooling communities.
- **Surprising comparative statics.** Reducing income inequality can increase consumption inequality if resources are moved away from the rich benefactor of a poor transfer component. Expanding the altruism network can also increase inequality, depending on which new link is strengthened.
- **Takeaway.** In altruistic networks, private transfers redistribute income, but not mechanically toward global equality. The path structure of transfers determines who is insured, who loses from policy changes, and whether altruism reduces or amplifies inequality.

2 Model setup (key objects)

2.1 Agents, incomes, and transfers

- There are $n \geq 2$ agents.
- Agent i has initial income $y_i^0 > 0$.
- Agent i may give a nonnegative transfer $t_{ij} \geq 0$ to agent j.

- By convention, $t_{ii} = 0$.
- The matrix of bilateral transfers is denoted by T.

Final consumption after transfers is

$$y_i = y_i^0 - \sum_j t_{ij} + \sum_k t_{ki}. \tag{1}$$

Interpretation. Transfers only redistribute income. Outgoing transfers reduce the giver's consumption; incoming transfers increase the receiver's consumption. Aggregate income is conserved:

$$\sum_i y_i = \sum_i y_i^0.$$

2.2 Private utility

Each agent has private utility

$$u_i(y_i),$$

where u_i is increasing and strictly concave:

$$u_i'(y) > 0, \qquad u_i''(y) < 0, \qquad \lim_{y \to \infty} u_i'(y) = 0.$$

Interpretation. Agents like consumption, but marginal utility declines. This makes richer agents more willing to give to poorer agents they care about.

2.3 Primitive and reduced-form altruism

The paper begins with a general social-utility representation:

$$v_i(y) = u_i(y_i) + \sum_j a_{ij} u_j(y_j) + \sum_j b_{ij} v_j(y). \tag{2}$$

Here:

- a_{ij} captures direct concern for j's private utility;
- b_{ij} captures concern for j's social utility;
- the social utilities v_i are jointly determined.

When the system is well behaved, this can be written in reduced form as

$$v_i(y) = u_i(y_i) + \sum_j \alpha_{ij} u_j(y_j), \tag{3}$$

where $\alpha_{ij} \geq 0$ is the reduced-form altruism weight.

Interpretation. The reduced-form network α is the object that matters for transfers. If $\alpha_{ij} > 0$, agent i ultimately cares about agent j's private well-being. The stronger α_{ij} is, the more willing i is to give to j.

2.4 The altruism network

The collection of coefficients α_{ij} defines the altruism network.

- $\alpha_{ij} = 0$: i does not care about j enough to give directly.
- $0 < \alpha_{ij} < 1$: i values j's private utility, but less than her own.
- α_{ij} need not be symmetric: i can care about j more than j cares about i.

The paper assumes

$$u_i'(y) > \alpha_{ij} u_j'(y) \qquad \text{for all } i, j, y. \tag{4}$$

Interpretation. Agent i always values one more unit for herself more than one more unit for j when both have the same consumption. Therefore, a giver never makes the receiver richer than herself in equilibrium.

2.5 Virtual transfer costs

For every altruistic link $\alpha_{ij} > 0$, define

$$c_{ij} = -\ln(\alpha_{ij}).$$

Interpretation. This is not a technological cost. It is a shadow or virtual cost induced by weaker altruism. A larger α_{ij} means stronger altruism and therefore a lower virtual cost c_{ij}.

A path has cost equal to the sum of link costs along the path. Because costs are $-\ln(\alpha_{ij})$, minimizing path cost is equivalent to maximizing the product of altruism weights along the path.

3 Equilibrium analysis (all equations + interpretation)

3.1 Individual best responses

Given transfers by all other agents, agent i chooses outgoing transfers t_{ij} to maximize her social utility.

Because utility is concave, the first-order conditions characterize best responses. A transfer network T is a Nash equilibrium if and only if

$$u_i'(y_i) \geq \alpha_{ij} u_j'(y_j), \qquad t_{ij} > 0 \Rightarrow u_i'(y_i) = \alpha_{ij} u_j'(y_j). \tag{5}$$

Read equation (5).

- If i does not give to j, then i's marginal utility from keeping a dollar is at least as high as her altruistically discounted marginal utility from giving that dollar to j.
- If i gives to j, the two marginal values are equal.
- If $\alpha_{ij} = 0$, then i never gives to j.

Economic intuition. Transfers equalize marginal utility only after discounting the receiver's marginal utility by altruism. A strongly altruistic link allows a larger transfer; a weak link requires the receiver to be much poorer before giving is worthwhile.

3.2 CARA example

Suppose agents have homogeneous CARA utility

$$u_i(y) = -\frac{e^{-Ay}}{A}.$$

Then equation (5) implies

$$t_{ij} > 0 \quad \Rightarrow \quad y_i = y_j + \frac{c_{ij}}{A}.$$

Interpretation. Along an active transfer link, the giver remains richer than the receiver by an amount proportional to the virtual transfer cost. A stronger altruistic link has lower c_{ij}, so it allows more equal consumption between giver and receiver.

3.3 Potential function

The paper's key technical step is to show that Nash equilibria maximize the concave potential

$$\phi(T) = \sum_i \int^{y_i} \ln\left(u_i'(x)\right) dx - \sum_{i,j:\,\alpha_{ij}>0} c_{ij} t_{ij}. \tag{6}$$

Taking the derivative with respect to t_{ij} gives

$$\frac{\partial \phi}{\partial t_{ij}} = -\ln\left(u_i'(y_i)\right) + \ln\left(u_j'(y_j)\right) + \ln(\alpha_{ij}).$$

Thus the first-order condition for maximizing ϕ is exactly the equilibrium condition in equation (5).

Interpretation. Even though each agent chooses transfers noncooperatively, the equilibrium can be found as if society were maximizing one common objective. This does not mean equilibrium is Pareto efficient. The potential includes virtual transfer costs, so it differs from utilitarian welfare.

3.4 Minimum-cost-flow interpretation

The potential can be read as

$$\text{benefit from smoother consumption} - \text{virtual cost of moving resources through altruistic links.}$$

Therefore, holding the final consumption vector fixed, equilibrium transfers minimize the virtual cost of implementing that redistribution. This creates the minimum-cost-flow result:

- money flows through least-cost paths;
- cycles are wasteful, because eliminating a cycle saves cost without changing consumption;
- direct transfers are not used if an indirect route is stronger.

Important implication. If the direct link $i \to j$ is weaker than an indirect path from i to j, then money will not flow directly from i to j in equilibrium.

3.5 Theorem 1: existence, uniqueness, and geometry

Theorem 1. A transfer network T is a Nash equilibrium if and only if it maximizes the concave potential ϕ over feasible transfer networks. A Nash equilibrium exists. Equilibrium transfers are acyclic and flow through least-cost paths of the altruism network. Equilibrium consumption is unique, continuous in income and altruism weights, and second-order stochastically dominates the initial income profile. Generically, equilibrium transfers are unique and form a forest.

Read.

- **Existence:** the transfer game always has an equilibrium.
- **Unique consumption:** even if multiple transfer networks can implement the same outcome, the final consumption vector is pinned down.
- **Acyclic transfers:** money does not circulate in loops.
- **Forest structure:** generically, the transfer network has no undirected cycles.
- **Redistribution:** equilibrium consumption is less unequal than income in the sense of second-order stochastic dominance.

Why this matters. The theorem makes the model operational. It tells us that transfer patterns are not arbitrary: they have strong graph-theoretic restrictions that can be tested in transfer data.

3.6 Transitive altruism and transfer intermediaries

The paper defines the transitive closure of the altruism network. Intuitively, $\widehat{\alpha}_{ij}$ captures the strongest indirect connection from i to j.

If

$$\alpha_{ij} < \widehat{\alpha}_{ij},$$

then the direct link from i to j is weaker than an indirect route, so the direct transfer t_{ij} is never used.

A **transfer intermediary** is an agent who both receives and gives in equilibrium.

Interpretation. Intermediaries arise when money can move more effectively through a chain of stronger altruistic links than through a weak direct link. They transmit support from richer parts of the network to poorer parts of the network.

3.7 Theorem 2: when intermediaries can be avoided

Theorem 2. There exists a Nash equilibrium without transfer intermediaries for every initial income profile if and only if the altruism network is transitive. This condition holds whenever altruism is consistent with deferential caring.

Read.

- If altruism is transitive, indirect concern can always be replaced by direct concern. Intermediaries are not needed.
- If altruism is not transitive, intermediaries may be essential.

- If agents care about others' social utilities rather than only private utilities, the reduced-form altruism network becomes transitive.

Economic intuition. Suppose i cares about j, and j cares about k. If i cares about j's *social* utility, then i indirectly cares about k too. In reduced form, this creates a direct concern from i to k. This is why deferential caring rules out the need for intermediaries.

4 Comparative statics (income shocks, redistribution, and altruism shocks)

4.1 Positive income shocks

The first comparative-static result is monotonicity: increasing one agent's income cannot hurt anyone's equilibrium consumption.

Intuition. A positive shock injects resources into the transfer system. Private transfers adjust, but because no money leaves the economy and equilibrium consumption is uniquely determined, everyone weakly benefits.

4.2 Theorem 3: income-pooling components

Let C_i denote the connected component of agent i in the initial equilibrium transfer network, using undirected connections. Let

$$y^0(C_i) = \sum_{k \in C_i} y_k^0$$

be the aggregate initial income of that component.

Theorem 3. Equilibrium consumption y_i is increasing in own income and weakly increasing in other agents' incomes. Generically, around a given income profile, there exists an increasing function f_i such that

$$y_i = f_i(y^0(C_i)).$$

Read.

- Locally, only the aggregate income of i's transfer component matters for i's consumption.
- Redistributing income within a transfer component is neutral.
- Redistributing income across transfer components is not neutral.

Interpretation. Each connected component of the equilibrium transfer network behaves like an income-pooling community. But the communities are determined by actual equilibrium transfers, not by the underlying altruism network.

4.3 Three-agent chain example

Suppose $C = \{i, j, k\}$, i gives to j, and j gives to k. Under common CARA utilities with $A = 1$, equilibrium satisfies

$$y_i = y_j + c_{ij}, \qquad y_j = y_k + c_{jk}, \qquad y_i + y_j + y_k = y^0(C).$$

Solving gives

$$y_i = \frac{1}{3}y^0(C) + \frac{2}{3}c_{ij} + \frac{1}{3}c_{jk},$$

$$y_j = \frac{1}{3}y^0(C) - \frac{1}{3}c_{ij} + \frac{1}{3}c_{jk}, \qquad y_k = \frac{1}{3}y^0(C) - \frac{1}{3}c_{ij} - \frac{2}{3}c_{jk}.$$

Read. All three consumptions move with the component's total income. Internal redistribution within C does not change any of the three consumption levels, as long as the transfer structure is unchanged.

4.4 Why reducing income inequality can increase consumption inequality

A standard intuition says that transferring resources from rich to poor should reduce inequality. The paper shows that this is not always true once private transfers operate through networks.

Mechanism.

- A poor agent may be supported by a richer agent in the same transfer component.
- If policy takes income away from that rich supporter, the whole component becomes poorer.
- Even if the policy gives money to a poorer agent elsewhere, the poorest supported agent may lose because her component's aggregate income falls.

Takeaway. Public redistribution interacts with private redistribution. A policy that reduces inequality in initial incomes can increase inequality in final consumption.

4.5 Theorem 4: changing one altruistic link

Now consider increasing one altruism weight α_{ij}. This makes agent i more altruistic toward agent j.

Let $S_i(T)$ and $S_j(T)$ be the two subcomponents obtained after removing the active transfer link $i \to j$ from the initial transfer network.

Theorem 4. Generically, a small effective increase in α_{ij} has the following effect:

$$S_i(T) = \{k : \widetilde{y}_k < y_k\}, \qquad S_j(T) = \{k : \widetilde{y}_k > y_k\}.$$

For a larger effective increase, the analogous loss and gain statements hold for the subcomponents that remain connected through links active both before and after the change.

Read.

- The giver side of the transfer link loses consumption.
- The receiver side gains consumption.
- The effect propagates through the existing transfer network, not merely across the changed altruistic link.

Economic intuition. If i becomes more willing to give to j, then j and the agents downstream from j gain. Agents upstream or connected to i may lose because more resources are redirected away from them.

4.6 Why expanding altruism can increase inequality

Adding or strengthening an altruistic link sounds like it should make the society more equal. But the paper shows that the effect depends on where the link is added.

- If the new link redirects resources toward poorer agents, inequality can fall.
- If the new link drains resources from a poor component or benefits agents who are already relatively better off, inequality can rise.
- The sign is determined by the initial transfer network and the identity of the affected subcomponents.

Takeaway. More altruism is not automatically more equality. Network position matters.

5 Empirical implications and limitations (what would make the model testable)

5.1 Observable implications

The theory generates several testable predictions for data on private transfers:

- **Acyclicity:** equilibrium transfers should not contain directed cycles.
- **Forest structure:** generically, the transfer graph should not contain undirected cycles.
- **Least-cost paths:** observed transfers reveal information about the relative strength of altruistic links.
- **Intermediaries:** observing a person who both receives and gives can indicate intransitive reduced-form altruism.
- **Neutrality failures:** redistributions across transfer components should affect consumption, while small redistributions within a component should be offset by private transfers.

5.2 What the model abstracts from

- Transfers are static rather than dynamic.
- Altruism weights are taken as fixed.
- There are no explicit transaction costs, enforcement problems, or information frictions.
- The model focuses on altruistic motives, not exchange, reciprocity, warm-glow giving, paternalism, or strategic family bargaining.
- The framework predicts final consumption and transfers, but measuring private utility and altruism weights in data remains difficult.

Interpretation. The paper is a benchmark theory. Its value is not that it captures every motive for transfers, but that it shows how pure altruism behaves once family and friendship links form a network.

6 Figures and tables (original titles from the paper)

The uploaded paper contains three figures and no tables. Below I keep the paper's original figure numbers and titles. Adjacent figures are placed two side-by-side where possible.

6.1 Visual guide

Figure 1.—The graph of transfers with a rich benefactor.

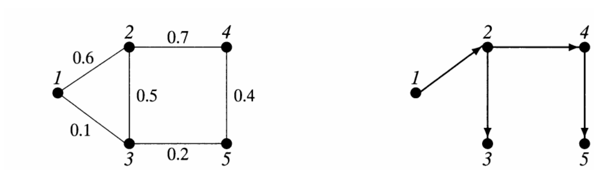
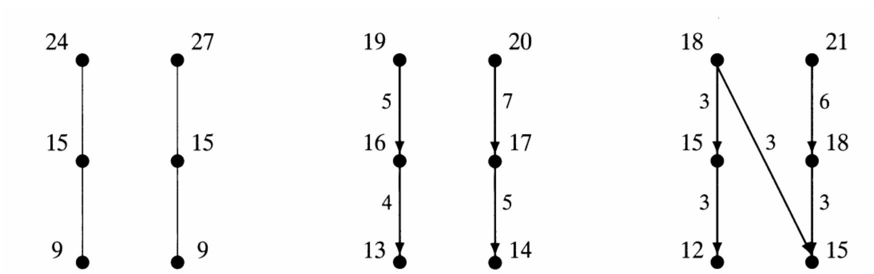


Figure 2.—Reducing income inequality may increase consumption inequality.

Figure 3.—An expansion of the altruism network can increase inequality.



6.2 Figure 1.—The graph of transfers with a rich benefactor.

Read. The left panel shows the altruism network with link strengths. The right panel shows the equilibrium transfer graph when agent 1 is a rich benefactor.

Key point. Money does not necessarily flow through every direct altruistic link. In the example, the direct link from 1 to 3 is weak relative to the indirect route through 2, so support flows through the stronger path.

Why it matters. The figure visualizes the least-cost-path property. Transfer intermediaries arise naturally when indirect altruistic connections are stronger than direct ones.

6.3 Figure 2.—Reducing income inequality may increase consumption inequality.

Read. The figure compares initial incomes and final consumption before and after a redistribution. The redistribution reduces inequality in initial incomes but worsens inequality in final consumption.

Key point. The reason is component-level income pooling. If redistribution removes resources from the rich supporter of a poor component, the supported poor agent can become worse off even though policy appears redistributive in the aggregate.

Why it matters. The figure shows that public redistribution and private transfers are not separable. Policies are filtered through the equilibrium transfer network.

6.4 Figure 3.—An expansion of the altruism network can increase inequality.

Read. The left panel shows the original altruism network. The middle panel shows equilibrium transfers and consumption. The right panel shows what happens after a new altruistic connection is added.

Key point. The new connection shifts consumption toward agents on one side of the transfer network and away from agents on the other side. The result can be higher inequality.

Why it matters. Network expansion is not automatically welfare-improving or inequality-reducing. Which link expands is crucial.

7 Short summary

- Agents embedded in a fixed altruism network make bilateral transfers to poorer people they care about.
- Nash equilibria solve a concave potential maximization problem.
- Transfers flow through least-cost paths of the altruism network and are acyclic; generically, the transfer graph is a forest.
- Equilibrium consumption is unique, even when transfer patterns are not.
- Transfer intermediaries arise when the altruism network is intransitive.
- Small redistributions are neutral only within components of the equilibrium transfer network.
- Reducing income inequality can increase consumption inequality.
- Strengthening altruism can also increase inequality if it redirects resources in the wrong part of the network.

8 Conclusion

Core conclusion. Altruistic transfers in networks are governed by the geometry of the altruism network. The equilibrium transfer graph is not simply a list of who cares about whom; it is a cost-minimizing structure that sends money through the strongest altruistic paths.

Economic intuition. Private transfers create local income-pooling groups. These groups are endogenous: they are the connected components of the equilibrium transfer network. A policy or social change affects an agent through the component in which the agent is pooled, not only through the agent's own income or direct links.

Takeaway. Networks fundamentally change the economics of altruism. They create intermediaries, partial neutrality, indirect propagation of shocks, and counterintuitive inequality effects. To understand private redistribution, we need to know not only who is rich and who is poor, but also how altruistic links connect them.